

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
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Section / Batch:	BE-CSE(AI)	Date:	24/07/2026

Code of Conduct

I ____K.GNANESWAR KUMAR(192572105)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: k.gnaneswar kumar____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

PSEUDOCODE WRITING TASK OPEN

CO5_AT1_Pseudocode Writing Task

 0/5 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

 10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

 10/10 answered

Review →

CO3_AT3_MCQ OPEN

Multiple Choice Questions - Non Linear Data Structures

 0/25 answered

Attempt →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int binarySearch(int arr[],int n,int key)  
3 {  
4     int low=0,high=n-1;  
5     while(low<=high)  
6     {  
7         int mid=(low+high)/2;  
8         if(arr[mid]==key)  
9             return mid;  
10        else if(arr[mid]<key)  
11            low=mid+1;  
12        else  
13            high=mid-1;  
14    }  
15    return -1;  
16 }  
17 int main()  
18 {  
19     int n,key;  
20     scanf("%d",&n);  
21     int arr[n];  
22     for(int i=0;i<n;i++)  
23         scanf("%d",&arr[i]);  
24     scanf("%d",&key);  
25     printf("%d",binarySearch(arr,n,key));  
26     return 0;  
27 }
```

Submitted

Test Input

```
5  
10 20 30 40 50  
30
```

Output

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int stack[100];  
3 int top=-1;  
4 int pop()  
5 {  
6     if(top==0)  
7     {  
8         printf("underflow");  
9         return -1;  
10    }  
11    return stack[top--];  
12 }  
13 int main()  
14 {  
15     stack[++top]=10;  
16     stack[++top]=20;  
17     stack[++top]=30;  
18     printf("%d",pop());  
19     return 0;  
20 }
```

Submitted

Test Input

3
10 20 30

Output

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int stack[5];  
3 int top;  
4 void init()  
5 {  
6     top=-1;  
7 }  
8 int main()  
9 {  
10     init();  
11     printf("%d",top);  
12     return 0;  
13 }
```

Test Input

stdin input (optional)



Output

-1

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){
    while(stack[top] != ' '){
        postfix[j++] = pop();
    }
    pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 char stack[100];
3 char postfix[100];
4 int top=-1;
5 int i=0;
6 void push(char ch)
7 {
8     stack[++top]=ch;
9 }
10 char pop()
11 {
12     if(top==1)
13         return'\0';
14     return stack[top--];
15 }
16 int main()
17 {
18     int n;
19     char ch;
20     scanf("%d",&n);
21     for(int j=0;j<n;j++)
22     {
23         scanf("%c",&ch);
24         push(ch);
25     }
26     scanf("%c",&ch);
27     if(ch==' '){
28         while(top!=-1&&stack[top]!=' '){
29             postfix[i++]=pop();
30         }
31         if(top!=-1)
32             pop();
33     }
34     postfix[i]='\0';
35     printf("%s",postfix);
36     return 0;
37 }
```

Test Input

3

(

+

+

Output

++

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

05 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if(ch == '(' && stack[top] == '(') ||  
(ch == '[' && stack[top] == '[') {  
    pop();  
} else {  
    return 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[100];  
4 int top = -1;  
5  
6 void push(char ch)  
7 {  
8     stack[++top] = ch;  
9 }  
10  
11 char pop()  
12 {  
13     return stack[top--];  
14 }  
15  
16 int main()  
17 {  
18     int n;  
19     char ch;  
20  
21     scanf("%d", &n);  
22  
23     for(int i = 0; i < n; i++)  
24     {  
25         scanf("%c", &ch);  
26  
27         if(ch == '(' || ch == '[' || ch == '{')  
28         {  
29             push(ch);  
30         }  
31         else  
32         {  
33             if((ch == ')' && stack[top] == '(') ||  
34                (ch == ']' && stack[top] == '[') ||  
35                (ch == '}' && stack[top] == '{'))  
36             {  
37                 pop();  
38             }  
39             else  
40             {  
41                 printf("Not Balanced");  
42                 return 0;  
43             }  
44         }  
45     }  
46  
47     if(top == -1)  
48         printf("Balanced");  
49     else  
50         printf("Not Balanced");  
51  
52     return 0;  
53 }
```

Submitted

Test Input

```
5  
(  
[  
{  
}
```

Output

balanced

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
3 marks

Q2

Stack
3 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
3 marks

Q6

Singly Linked List
3 marks

Q7

Stack
3 marks

Q8

Linked List
3 marks

Q9

Linked List
3 marks

Q10

Queue
5 marks

10/10 answered

06 Singly Linked List

3 marks

What is the issue?

```
temp = head;
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 };
9
10 int main()
11 {
12     struct Node *head = NULL, *temp = NULL, *newNode = NULL;
13     int n, value;
14
15     scanf("%d", &n);
16
17     for(int i = 0; i < n; i++)
18     {
19         scanf("%d", &value);
20
21         newNode = (struct Node *)malloc(sizeof(struct Node));
22         newNode->data = value;
23         newNode->next = NULL;
24
25         if(head == NULL)
26         {
27             head = newNode;
28             temp = head;
29         }
30         else
31         {
32             temp->next = newNode;
33             temp = newNode;
34         }
35     }
36
37     temp = head;
38
39     while(temp != NULL)
40     {
41         printf("%d ", temp->data);
42         temp = temp->next;
43     }
44
45     return 0;
46 }
```

Test Input

```
5
10 20 30 40 50
```

Output

```
10 20 30 40 50
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search
1 marks

Q2

Stack
1 marks

Q3

Stack
1 marks

Q4

Stack
1 marks

Q5

Stack - Identifying Symbols
1 marks

Q6

Simply Linked List
1 marks

Q7

Stack
1 marks

Q8

Linked List
1 marks

Q9

Linked List
1 marks

Q10

Queue
1 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack);  
if top == 0;  
print("Underflow")  
else:  
return stack[top-1]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[100];  
4 int top = -1;  
5  
6 void push(int value)  
7 {  
8     stack[++top] = value;  
9 }  
10  
11 int pop()  
12 {  
13     if(top == -1)  
14     {  
15         printf("Underflow");  
16         return -1;  
17     }  
18     else  
19     {  
20         return stack[top--];  
21     }  
22 }  
23  
24 int main()  
25 {  
26     int n, value;  
27  
28     scanf("%d", &n);  
29  
30     for(int i = 0; i < n; i++)  
31     {  
32         scanf("%d", &value);  
33         push(value);  
34     }  
35  
36     printf("%d", pop());  
37  
38     return 0;  
39 }
```

Test Input

```
1  
10 20 30 40 50
```

Output

```
30
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Words Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

08. Linked List

5 marks

Identify the errors in the following code.

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 };
9
10 int main()
11 {
12     struct Node *head = NULL, *temp = NULL, *newNode;
13     struct Node *prev = NULL, *current, *next;
14
15     int n, value;
16
17     scanf("%d", &n);
18
19     for(int i = 0; i < n; i++)
20     {
21         scanf("%d", &value);
22
23         newNode = (struct Node *)malloc(sizeof(struct Node));
24         newNode->data = value;
25         newNode->next = NULL;
26
27         if(head == NULL)
28         {
29             head = newNode;
30             temp = head;
31         }
32         else
33         {
34             temp->next = newNode;
35             temp = newNode;
36         }
37     }
38
39     current = head;
40
41     while(current != NULL)
42     {
43         next = current->next;
44         current->next = prev;
45         prev = current;
46         current = next;
47     }
48
49     head = prev;
50
51     temp = head;
52     while(temp != NULL)
53     {
54         printf("%d ", temp->data);
55         temp = temp->next;
56     }
57
58     return 0;
59 }
```

Test Input

```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

🔄 Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly linked list
5 marksQ7
Stack
5 marksQ8
Linked list
5 marksQ9
Linked list
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

What is the error in the below code?

new.next = head
head = new

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 }
9
10 int main()
11 {
12     struct Node *head = NULL, *temp, *newNode;
13     int n, value;
14
15     scanf("%d", &n);
16
17     for(int i = 0; i < n; i++)
18     {
19         scanf("%d", &value);
20
21         newNode = (struct Node *)malloc(sizeof(struct Node));
22         newNode->data = value;
23
24         newNode->next = head;
25         head = newNode;
26     }
27
28     temp = head;
29
30     while(temp != NULL)
31     {
32         printf("%d ", temp->data);
33         temp = temp->next;
34     }
35
36     return 0;
37 }
```

Test Input

5
10 10 10 10 10

Output

10 10 10 10 10

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q10: Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
If front == -1:  
    print("Underflow")  
else:  
    return Q[front+1]
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int Q[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int value)  
9 {  
10     if(rear == MAX - 1)  
11     {  
12         printf("Overflow");  
13         return;  
14     }  
15     if(front == -1)  
16         front = 0;  
17     Q[++rear] = value;  
18 }  
19  
20 int dequeue()  
21 {  
22     if(front == -1)  
23     {  
24         printf("Underflow");  
25         return -1;  
26     }  
27     int item = Q[front];  
28     if(front == rear)  
29     {  
30         front = rear = -1;  
31     }  
32     else  
33     {  
34         front++;  
35     }  
36     return item;  
37 }  
38  
39 int main()  
40 {  
41     int n, value;  
42  
43     scanf("%d", &n);  
44     for(int i = 0; i < n; i++)  
45     {  
46         scanf("%d", &value);  
47         enqueue(value);  
48     }  
49     printf("%d", dequeue());  
50     return 0;  
51 }
```

Submitted

Test Input

5
10 20 30 40 50

Output

10

Pending manual review

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1

Stack
0 marks

✓

Q2

Pointer (Error)
0 marks

✓

Q3

Stack
0 marks

✓

Q4

Stack
0 marks

✓

Q5

Stack (stack not)
0 marks

✓

Q6

Stack
0 marks

✓

Q7

Stack - Output
0 marks

✓

Q8

Memory Address
0 marks

✓

Q9

Stack (Error)
0 marks

✓

Q10

Stack - Pointer Expression
0 marks

✓

10/10 answered

Stack

5 marks

Point out the error?

```
peek(stack);  
return stack[top+1];
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int aStack[100];  
4 int top = -1;  
5  
6 void push(int value)  
7 {  
8     aStack[++top] = value;  
9 }  
10  
11 int peek()  
12 {  
13     if(top == -1)  
14     {  
15         printf("Stack Underflow");  
16         return -1;  
17     }  
18     return aStack[top];  
19 }  
20  
21 int main()  
22 {  
23     int n, value;  
24     printf("N: ", 0);  
25     scanf("%d", &n);  
26     for(int i = 0; i < n; i++)  
27     {  
28         printf("N: ", value);  
29         push(value);  
30     }  
31     printf("N: ", peek());  
32     return 0;  
33 }
```

Submitted

Test Input

```
5  
10 20 30 40 50
```

Output

00

Pending manual review



CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1
Block
3 marksQ2
Function/Queue
3 marksQ3
Block
3 marksQ4
Block
3 marksQ5
Data type/linked list
3 marksQ6
Queue
3 marksQ7
Queue/Deque
3 marksQ8
Binary Search
3 marksQ9
Circular Queue
3 marksQ10
Stack/Priority Queue
3 marks

10/10 answered

Circular Queue

3 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;  
queue[rear] = vehicle;  
if(rear == front)  
    printf("Queue Full");  
}
```

★ Debugging Task: Find out for the bugs in the code above already. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 2  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int value)  
9 {  
10     if((rear + 1) % MAX == front)  
11     {  
12         printf("Queue Full");  
13         return;  
14     }  
15     if(front == -1)  
16         front = 0;  
17     rear = (rear + 1) % MAX;  
18     queue[rear] = value;  
19 }  
20  
21 int main()  
22 {  
23     int n, value;  
24     scanf("%d", &n);  
25  
26     for(int i = 0; i < n; i++)  
27     {  
28         scanf("%d", &value);  
29         enqueue(value);  
30     }  
31     printf("Queue: ");  
32  
33     for(int i = front; i < (rear + 1) % MAX; i = (i + 1) % MAX)  
34     {  
35         printf("%d ", queue[i]);  
36     }  
37     printf("\n");  
38     return 0;  
39 }
```

Test Input

5
10 20 30 40

Output

Queue: 10 20 30 40

Pending manual review

Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack
4 marks**Q2**
The size of array
4 marks**Q3**
Stack
4 marks**Q4**
Stack
4 marks**Q5**
Circularly linked list
4 marks**Q6**
Queue
4 marks**Q7**
Queue - Simple
4 marks**Q8**
Binary Search
4 marks**Q9**
Circular Queue
4 marks**Q10**
Stack - Recursive Implementation
4 marks

10/10 answered

Stack

5 marks

Point out the error:
if top == MAX
Overflow

Debugging Task: Find and fix the bugs in the code below. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Overflow");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int main()
19 {
20     int n, value;
21     scanf("%d", &n);
22     for(int i = 0; i < n; i++)
23     {
24         scanf("%d", &value);
25         push(value);
26     }
27     printf("Stack: ");
28     for(int i = 0; i < top; i++)
29     {
30         printf("%d ", stack[i]);
31     }
32     return 0;
33 }
```

Submitted

Test Input

5
10 20 30 40 50

Output

StackSize: 10 20 30 40 50

[Resolving manual review](#)

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)
[← Back](#)
QUESTIONS

- Q1
Stack
0 marks
- Q2
Priority Queue
0 marks
- Q3
Stack
0 marks
- Q4
Stack
0 marks
- Q5
Priority Queue List
0 marks
- Q6
Queue
0 marks
- Q7
Queue - Binary
0 marks
- Q8
Binary Search
0 marks
- Q9
Priority Queue
0 marks
- Q10
Stack - Binary Expression
0 marks

10/10 answered

Stack

0 marks

 Identify the issue in the below pseudocode.
 POP();
 if top == 0 -- Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 #define MAX 5
4
5 int aStack[100];
6 int top = -1;
7
8 void push(int value)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Overflow\n");
13         return;
14     }
15     aStack[++top] = value;
16 }
17
18 int pop()
19 {
20     if (top == -1)
21     {
22         printf("Underflow\n");
23         return -1;
24     }
25     return aStack[top--];
26 }
27
28 int main()
29 {
30     int n, value;
31     scanf("%d", &n);
32     for (int i = 0; i < n; i++)
33     {
34         scanf("%d", &value);
35         push(value);
36     }
37     printf("%d", pop());
38     return 0;
39 }

```

Submitted

Test Input

0

Output

Underflow

Pending manual review

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1

Node
0 marks

Q3

Doubly Linked List
0 marks

Q3

Node
0 marks

Q4

Node
0 marks

Q5

Doubly Linked List
0 marks

Q6

Queue
0 marks

Q7

Queue, Dequeue
0 marks

Q8

Binary Search
0 marks

Q9

Doubly Linked List
0 marks

Q10

Node, Print by Reversal
0 marks

13/12 answered

Q5: Doubly Linked List

Course

Identify the error in this DoublyLinkedList node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <iostream>  
2 #include <string>  
3  
4 struct Node  
5 {  
6     int data;  
7     struct Node *prev;  
8     struct Node *next;  
9 };  
10  
11 void deleteNode(struct Node *del)  
12 {  
13     if (del == NULL)  
14         return;  
15  
16     if (del->prev != NULL)  
17         del->prev->next = del->next;  
18  
19     if (del->next != NULL)  
20         del->next->prev = del->prev;  
21  
22     delete(del);  
23 }  
24  
25 int main()  
26 {  
27     struct Node *head = NULL, *temp = NULL, *newNode;  
28     int n, value, prev;  
29  
30     cout << "N: ";  
31  
32     for (int i = 0; i < n; i++)  
33     {  
34         cout << "N: ";  
35  
36         newNode = (struct Node *) malloc(sizeof(struct Node));  
37         newNode->data = value;  
38         newNode->prev = NULL;  
39         newNode->next = temp;  
40  
41         if (head == NULL)  
42             head = newNode;  
43         else  
44             temp->next = newNode;  
45  
46         temp = newNode;  
47     }  
48  
49     cout << "N: ";  
50  
51     temp = head;  
52     for (int i = 0; i < n; i++)  
53     {  
54         temp->prev = temp->next;  
55         temp = temp->prev;  
56     }  
57  
58     cout << "N: ";  
59  
60     temp = head;  
61     while (temp != NULL)  
62     {  
63         cout << "N: ";  
64         temp = temp->next;  
65     }  
66  
67     cout << "N: ";  
68  
69     temp = head;  
70     while (temp != NULL)  
71     {  
72         cout << "N: ";  
73         temp = temp->next;  
74     }  
75  
76     cout << "N: ";  
77  
78     temp = head;  
79     while (temp != NULL)  
80     {  
81         cout << "N: ";  
82         temp = temp->next;  
83     }  
84  
85     cout << "N: ";  
86  
87     temp = head;  
88     while (temp != NULL)  
89     {  
90         cout << "N: ";  
91         temp = temp->next;  
92     }  
93  
94     cout << "N: ";  
95  
96     temp = head;  
97     while (temp != NULL)  
98     {  
99         cout << "N: ";  
100        temp = temp->next;  
101    }  
102  
103    cout << "N: ";  
104  
105    temp = head;  
106    while (temp != NULL)  
107    {  
108        cout << "N: ";  
109        temp = temp->next;  
110    }  
111  
112    cout << "N: ";  
113  
114    temp = head;  
115    while (temp != NULL)  
116    {  
117        cout << "N: ";  
118        temp = temp->next;  
119    }  
120  
121    cout << "N: ";  
122  
123    temp = head;  
124    while (temp != NULL)  
125    {  
126        cout << "N: ";  
127        temp = temp->next;  
128    }  
129  
130    cout << "N: ";  
131  
132    temp = head;  
133    while (temp != NULL)  
134    {  
135        cout << "N: ";  
136        temp = temp->next;  
137    }  
138  
139    cout << "N: ";  
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141    temp = head;  
142    while (temp != NULL)  
143    {  
144        cout << "N: ";  
145        temp = temp->next;  
146    }  
147  
148    cout << "N: ";  
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150    temp = head;  
151    while (temp != NULL)  
152    {  
153        cout << "N: ";  
154        temp = temp->next;  
155    }  
156  
157    cout << "N: ";  
158  
159    temp = head;  
160    while (temp != NULL)  
161    {  
162        cout << "N: ";  
163        temp = temp->next;  
164    }  
165  
166    cout << "N: ";  
167  
168    temp = head;  
169    while (temp != NULL)  
170    {  
171        cout << "N: ";  
172        temp = temp->next;  
173    }  
174  
175    cout << "N: ";  
176  
177    temp = head;  
178    while (temp != NULL)  
179    {  
180        cout << "N: ";  
181        temp = temp->next;  
182    }  
183  
184    cout << "N: ";  
185  
186    temp = head;  
187    while (temp != NULL)  
188    {  
189        cout << "N: ";  
190        temp = temp->next;  
191    }  
192  
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195    temp = head;  
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198        cout << "N: ";  
199        temp = temp->next;  
200    }  
201  
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204    temp = head;  
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206    {  
207        cout << "N: ";  
208        temp = temp->next;  
209    }  
210  
211    cout << "N: ";  
212  
213    temp = head;  
214    while (temp != NULL)  
215    {  
216        cout << "N: ";  
217        temp = temp->next;  
218    }  
219  
220    cout << "N: ";  
221  
222    temp = head;  
223    while (temp != NULL)  
224    {  
225        cout << "N: ";  
226        temp = temp->next;  
227    }  
228  
229    cout << "N: ";  
230  
231    temp = head;  
232    while (temp != NULL)  
233    {  
234        cout << "N: ";  
235        temp = temp->next;  
236    }  
237  
238    cout << "N: ";  
239  
240    temp = head;  
241    while (temp != NULL)  
242    {  
243        cout << "N: ";  
244        temp = temp->next;  
245    }  
246  
247    cout << "N: ";  
248  
249    temp = head;  
250    while (temp != NULL)  
251    {  
252        cout << "N: ";  
253        temp = temp->next;  
254    }  
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270        cout << "N: ";  
271        temp = temp->next;  
272    }  
273  
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278    {  
279        cout << "N: ";  
280        temp = temp->next;  
281    }  
282  
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533    }  
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536  
537    temp = head;  
538    while (temp != NULL)  
539    {  
540        cout << "N: ";  
541        temp = temp->next;  
542    }  
543  
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802        temp = temp->next;  
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808    while (temp != NULL)  
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816    temp = head;  
817    while (temp != NULL)  
818    {  
819        cout << "N: ";  
820        temp = temp->next;  
821    }  
822  
823    cout << "N: ";  
824  
825    temp = head;  
826    while (temp != NULL)  
827    {  
828        cout << "N: ";  
829        temp = temp->next;  
830    }  
831  
832    cout << "N: ";  
833  
834    temp = head;  
835    while (temp != NULL)  
836    {  
837        cout << "N: ";  
838        temp = temp->next;  
839    }  
840  
841    cout << "N: ";  
842  
843    temp = head;  
844    while (temp != NULL)  
845    {  
846        cout << "N: ";  
847        temp = temp->next;  
848    }  
849  
850    cout << "N: ";  
851  
852    temp = head;  
853    while (temp != NULL)  
854    {  
855        cout << "N: ";  
856        temp = temp->next;  
857    }  
858  
859    cout << "N: ";  
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861    temp = head;  
862    while (temp != NULL)  
863    {  
864        cout << "N: ";  
865        temp = temp->next;  
866    }  
867  
868    cout << "N: ";  
869  
870    temp = head;  
871    while (temp != NULL)  
872    {  
873        cout << "N: ";  
874        temp = temp->next;  
875    }  
876  
877    cout << "N: ";  
878  
879    temp = head;  
880    while (temp != NULL)  
881    {  
882        cout << "N: ";  
883        temp = temp->next;  
884    }  
885  
886    cout << "N: ";  
887  
888    temp = head;  
889    while (temp != NULL)  
890    {  
891        cout << "N: ";  
892        temp = temp->next;  
893    }  
894  
895    cout << "N: ";  
896  
897    temp = head;  
898    while (temp != NULL)  
899    {  
900        cout << "N: ";  
901        temp = temp->next;  
902    }  
903  
904    cout << "N: ";  
905  
906    temp = head;  
907    while (temp != NULL)  
908    {  
909        cout << "N: ";  
910        temp = temp->next;  
911    }  
912  
913    cout << "N: ";  
914  
915    temp = head;  
916    while (temp != NULL)  
917    {  
918        cout << "N: ";  
919        temp = temp->next;  
920    }  
921  
922    cout << "N: ";  
923  
924    temp = head;  
925    while (temp != NULL)  
926    {  
927        cout << "N: ";  
928        temp = temp->next;  
929    }  
930  
931    cout << "N: ";  
932  
933    temp = head;  
934    while (temp != NULL)  
935    {  
936        cout << "N: ";  
937        temp = temp->next;  
938    }  
939  
940    cout << "N: ";  
941  
942    temp = head;  
943    while (temp != NULL)  
944    {  
945        cout << "N: ";  
946        temp = temp->next;  
947    }  
948  
949    cout << "N: ";  
950  
951    temp = head;  
952    while (temp != NULL)  
953    {  
954        cout << "N: ";  
955        temp = temp->next;  
956    }  
957  
958    cout << "N: ";  
959  
960    temp = head;  
961    while (temp != NULL)  
962    {  
963        cout << "N: ";  
964        temp = temp->next;  
965    }  
966  
967    cout << "N: ";  
968  
969    temp = head;  
970    while (temp != NULL)  
971    {  
972        cout << "N: ";  
973        temp = temp->next;  
974    }  
975  
976    cout << "N: ";  
977  
978    temp = head;  
979    while (temp != NULL)  
980    {  
981        cout << "N: ";  
982        temp = temp->next;  
983    }  
984  
985    cout << "N: ";  
986  
987    temp = head;  
988    while (temp != NULL)  
989    {  
990        cout << "N: ";  
991        temp = temp->next;  
992    }  
993  
994    cout << "N: ";  
995  
996    temp = head;  
997    while (temp != NULL)  
998    {  
999        cout << "N: ";  
1000        temp = temp->next;  
1001    }  
1002  
1003    cout << "N: ";  
1004  
1005    temp = head;  
1006    while (temp != NULL)  
1007    {  
1008        cout << "N: ";  
1009        temp = temp->next;  
1010    }  
1011  
1012    cout << "N: ";  
1013  
1014    temp = head;  
1015    while (temp != NULL)  
1016    {  
1017        cout << "N: ";  
1018        temp = temp->next;  
1019    }  
1020  
1021    cout << "N: ";  
1022  
1023    temp = head;  
1024    while (temp != NULL)  
1025    {  
1026        cout << "N: ";  
1027        temp = temp->next;  
1028    }  
1029  
1030    cout << "N: ";  
1031  
1032    temp = head;  
1033    while (temp != NULL)  
1034    {  
1035        cout << "N: ";  
1036        temp = temp->next;  
1037    }  
1038  
1039    cout << "N: ";  
1040  
1041    temp = head;  
1042    while (temp != NULL
```

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

4/5 Back

QUESTIONS

Q1

Stack
5 marks

✓

Q2

Circular Queue
5 marks

✓

Q3

Stack
5 marks

✓

Q4

Stack
5 marks

✓

Q5

Circular Linked List
5 marks

✓

Q6

Queue
5 marks

✓

Q7

Circular Queue
5 marks

✓

Q8

Binary Search
5 marks

✓

Q9

Circular Queue
5 marks

✓

Q10

Stack - Prefix Expression
5 marks

✓

11/10 answered

Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data) {  
9     if (rear == MAX - 1) {  
10        printf("Overflow");  
11        return;  
12    }  
13    if (front == -1) {  
14        front = 0;  
15    }  
16    queue[++rear] = data;  
17 }  
18  
19 int main() {  
20     int n, data;  
21     scanf("%d", &n);  
22     for (int i = 0; i < n; i++) {  
23         scanf("%d", &data);  
24         enqueue(data);  
25     }  
26     printf("Queue: ");  
27     for (int i = front; i < rear; i++) {  
28         printf("%d ", queue[i]);  
29     }  
30     printf("\n");  
31     return 0;  
32 }
```

Test Input

5
10 20 30 40 50

Output

Queue: 10 20 30 40 50

Feeling stuck? Ask for help

Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1
Stack
4 marks
- Q2
Linked Queue
4 marks
- Q3
Stack
4 marks
- Q4
Stack
4 marks
- Q5
Priority Queue List
4 marks
- Q6
Queue
4 marks
- Q7
Queue - Deque
4 marks
- Q8
Binary Search
4 marks
- Q9
Priority Queue
4 marks
- Q10
Stack - Push & Pop operation
4 marks

10/10 answered

Q7 Queue - Deque

4 marks

Find the error in this Deque insertFront operation.

```
void insertFront(data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 1  
4  
5 int deque[MAX];  
6 int front = -1, rear = -1;  
7  
8 void insertFront(int data)  
9 {  
10     if ((front == 0 && rear == MAX - 1) ||  
11         front == rear + 1)  
12     {  
13         printf("Overflow");  
14         return;  
15     }  
16     if (front == -1)  
17     {  
18         front = rear = 0;  
19     }  
20     else if (front == 0)  
21     {  
22         front = MAX - 1;  
23     }  
24     else  
25     {  
26         front--;  
27     }  
28  
29     deque[front] = data;  
30 }  
31  
32 int main()  
33 {  
34     int n, data;  
35     scanf("%d", &n);  
36  
37     for (int i = 0; i < n; i++)  
38     {  
39         scanf("%d", &data);  
40         insertFront(data);  
41     }  
42  
43     printf("Deque: ");  
44  
45     int i = front;  
46     while (i != rear + 1)  
47     {  
48         printf("%d ", deque[i]);  
49  
50         if (i == rear)  
51             break;  
52  
53         i = (i + 1) % MAX;  
54     }  
55  
56     return 0;  
57 }
```

Test Input

```
5  
10 20 30 40 50
```

Output

Deque: 10 20 30 40 50

Pending manual review

Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Quick
5 marksQ2
Procedural Question
5 marksQ3
Quick
5 marksQ4
Quick
5 marksQ5
Complex Problem Solving
5 marksQ6
Question
5 marksQ7
Question - Simple
5 marksQ8
Binary Search
5 marksQ9
Procedural Question
5 marksQ10
Quick - Best for Performance
5 marks

0/10 answered

Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <iostream>  
2  
3 int binarySearch(int arr[], int l, int r, int x) {  
4     if (l > r)  
5         return -1;  
6  
7     int mid = l + (r - l) / 2;  
8  
9     if (arr[mid] == x)  
10        return mid;  
11  
12    if (arr[mid] > x)  
13        return binarySearch(arr, l, mid - 1, x);  
14  
15    return binarySearch(arr, mid + 1, r, x);  
16 }  
17  
18 int main() {  
19     int n;  
20  
21    cout << "N: ";  
22  
23    int arr[n];  
24  
25    for (int i = 0; i < n; i++)  
26        cout << " ";  
27  
28    cout << "N: ";  
29  
30    getChar("N", binarySearch(arr, 0, n - 1, x));  
31  
32    return 0;  
33 }
```

Submitted

Test Input

```
5  
1 10 20 30 40 50  
20
```

Output

Feeling stuck? Review

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

8/10/25

QUESTIONS

Q1

Read
0 marks

Q2

Answer Question
0 marks

Q3

Read
0 marks

Q4

Read
0 marks

Q5

Reading Lockout List
0 marks

Q6

Answer
0 marks

Q7

Answer (Group)
0 marks

Q8

Reading Thread
0 marks

Q9

Answer Question
0 marks

Q10

Read - Read by Registrator
0 marks

10/10 answered

Circular Queue

3 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 //CircularQueue.cpp  
2  
3 #include <iostream>  
4  
5 #define MAX 10  
6  
7 int queue[MAX];  
8 int front = -1, rear = -1;  
9  
10 void enqueue(int data) {  
11     if ((rear + 1) % MAX == front) {  
12         cout << "Queue Full!\n";  
13         return;  
14     }  
15     if (rear == -1) {  
16         rear = rear + 1;  
17     } else {  
18         rear = (rear + 1) % MAX;  
19     }  
20     queue[rear] = data;  
21 }  
22  
23 int dequeue() {  
24     if (front == -1) {  
25         cout << "Queue Empty!\n";  
26         return -1;  
27     }  
28     int data = queue[front];  
29     if (front == rear) {  
30         front = rear = -1;  
31     } else {  
32         front = (front + 1) % MAX;  
33     }  
34     return data;  
35 }  
36  
37 int main() {  
38     int n, value;  
39     cout << "N: ";  
40     cin >> n;  
41     for (int i = 0; i < n; i++) {  
42         cout << "Value: ";  
43         cin >> value;  
44         enqueue(value);  
45     }  
46     cout << "N: ";  
47     cout << dequeue();  
48 }
```

Test Input

```
5  
10 20 30 40 50
```

Output

10

Feeding output review

Submitted